

A Coxeter-type bound for zero-fiber rings of quaternionic quotient singularities

Lien Cartaya

Joint work with Stephen Griffeth

CIME Coxeter School, Cetraro 2026

0. Main idea

Given a reflection group W , the number

$$g = \frac{2N}{n}$$

controls the size of zero-fiber rings beyond real and complex reflection groups.

Here

$$N = \#\{\text{reflections in } W\}, \quad n = \text{rank}(W).$$

1. General zero-fiber ring

Let $D = \mathbb{R}, \mathbb{C}$, or $\mathbb{H}$.

Let V be a finite-dimensional right D -vector space, and let

$$W \leq GL(V)$$

be a finite group.

The quotient map is

$$\pi : V_{\mathbb{C}} \longrightarrow V_{\mathbb{C}}/W = \operatorname{Spec} \mathbb{C}[V]^W.$$

$$\pi^{-1}(0) \subseteq V_{\mathbb{C}}$$

The zero-fiber ring is

$$\mathbb{C}[\pi^{-1}(0)] = \frac{\mathbb{C}[V]}{\langle f - f(0) : f \in \mathbb{C}[V]^W \rangle}.$$

We write

$$\delta(W) = \dim_{\mathbb{C}} \mathbb{C}[\pi^{-1}(0)].$$

- If $D = \mathbb{R}, \mathbb{C}$ and W is a reflection group, $\delta(W) = |W|$.

2. Quaternionic reflection groups

$$D = \mathbb{H} = \frac{\mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k}{(i^2 = j^2 = k^2 = ijk = -1)}$$

Let V be a right $\mathbb{H}$ -vector space and let $W \leq GL_{\mathbb{H}}(V)$.

An element $r \in W$ is a reflection if

$$\operatorname{codim}_{\mathbb{H}} \operatorname{Fix}_V(r) = 1.$$

The group W is a quaternionic reflection group if

$$W = \langle \text{its reflections} \rangle.$$

3. Conjecture

Given an irreducible quaternionic reflection group W

$\exists$ an irreducible $L \in A_W\text{-mod}$, where A_W is the symplectic reflection algebra of W s.t.

- $\dim_{\mathbb{C}} L = (g+1)^n$
- $\dim_{\mathbb{C}} L^W = 1$

That implies

$$\exists \mathbb{C}[\pi^{-1}(0)]_W \twoheadrightarrow \text{gr}(L)$$

Therefore,

$$\delta(W) \geq (g+1)^n.$$

4. Main theorem

The conjecture holds for W

- irreducible
- imprimitive
- and $\text{rank}(W) = n \geq 3$

The technical proof is for:

$$W_n(\Gamma, \Delta) = \Gamma^n \rtimes S_n,$$

$$\Gamma \leq \mathbb{H}^\times,$$

a non-trivial rank-one q. r.
group

(i.e. a Kleinian group)

It also holds if W is:

- irreducible
- and $n \geq 6$

5. The symmetric group

If $\Gamma = \{1\}$ then

$$\Gamma^{\times n} \rtimes S_n = S_n$$

$$S_n \curvearrowright \{x_1 + \dots + x_n = 0\} = V \subset \mathbb{H}^n$$

Here

$$(g+1)^n = (n+1)^{n-1} = \delta(S_n)$$

(by M. Haiman, 2000)

6. Result for rank-one

$$\Gamma = C_2 \leq \mathbb{H}^\times, \quad V = \mathbb{H} \cong \mathbb{C}^2.$$

$$-1 \cdot (x, y) = (-x, -y).$$

The zero-fiber ring is

$$\mathbb{C}[\pi^{-1}(0)] = \frac{\mathbb{C}[x, y]}{(x^2, xy, y^2)} = \text{span}_{\mathbb{C}}\{1, x, y\}$$

$$\implies \delta(C_2) = 3.$$

For $\Gamma = C_2$, we have

$$N = 1, \quad n = 1, \quad g = \frac{2N}{n} = 2.$$

Therefore

$$\delta(C_2) = 3 = g + 1.$$

Theorem

For $\Gamma \leq \mathbb{H}^\times$ non-trivial,

$$\delta(\Gamma) = 2|\Gamma| - 1 = g + 1.$$

7. Principal tool

The rank-one (for $\Gamma \leq \mathbb{H}^\times$) symplectic reflection algebra is

$$H_c(\Gamma, \mathbb{C}^2) = \frac{\mathbb{C}\langle x, y \rangle \rtimes \Gamma}{xy - yx - \sum_{i \in \text{Irr}(\Gamma)} c_i e_i}.$$

This is a deformation of $\mathbb{C}[x, y] \rtimes \Gamma$.

- Crawley-Boevey and Holland classify its finite-dimensional simple modules using the affine Dynkin diagram associated with Γ .
- A trick to reduce the general case to rank 1 (It worket!)

Thank you for your attention

References

- [1] L. Cartaya and S. Griffeth, *Zero fibers of quaternionic quotient singularities*, Rev. Mat. Iberoam., 41 (2025), no. 4, pp. 1461–1487.
- [2] W. Crawley-Boevey and M. P. Holland, *Noncommutative deformations of Kleinian singularities*, Duke Math. J. 92 (1998), no. 3, 605–635.
- [3] I. Gordon, *On the quotient ring by diagonal invariants*, Invent. Math. 153 (2003), no. 3, 503–518.